

Supplementary Material for

“Deep Unfolded Shifted Power Iteration-Based ISAC Beamforming for Sum Rate and CRLB Balancing”

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I. SYSTEM MODEL

The base station (BS) has L_t transmit antennas and L_r sensing receive antennas. It simultaneously serves K single-antenna communications users and senses Q point targets over T symbol intervals. The transmitted signal is

$$\mathbf{c}[t] = \mathbf{W}\mathbf{s}[t], \quad \mathbf{W} = [\mathbf{w}_1, \dots, \mathbf{w}_K] \in \mathbb{C}^{L_t \times K}, \quad (1)$$

where the data streams are mutually independent and normalized. Hence,

$$\mathbf{S}\mathbf{S}^H \simeq T\mathbf{I}_K, \quad \mathbf{S} = [\mathbf{s}[1], \dots, \mathbf{s}[T]]. \quad (2)$$

The noiseless sensing echo is

$$\mathbf{u}_s[t] = \mathbf{B}\mathbf{D}\mathbf{A}^H\mathbf{W}\mathbf{s}[t]. \quad (3)$$

Here,

$$\mathbf{A} = [\mathbf{a}(\theta_1, \phi_1), \dots, \mathbf{a}(\theta_Q, \phi_Q)] \in \mathbb{C}^{L_t \times Q}, \quad (4)$$

$$\mathbf{B} = [\mathbf{b}(\theta_1, \phi_1), \dots, \mathbf{b}(\theta_Q, \phi_Q)] \in \mathbb{C}^{L_r \times Q}, \quad (5)$$

$$\mathbf{D} = \text{diag}(\mathbf{d}), \quad \mathbf{d} = [d_1, \dots, d_Q]^T. \quad (6)$$

The observation is

$$\mathbf{y}_s[t] = \mathbf{u}_s[t] + \mathbf{n}_s[t], \quad \mathbf{n}_s[t] \sim \mathcal{CN}(\mathbf{0}, \sigma_s^2 \mathbf{I}_{L_r}). \quad (7)$$

The unknown real-valued target parameter vector is

$$\boldsymbol{\omega} = [\boldsymbol{\theta}^T, \boldsymbol{\phi}^T, \text{Re}\{\mathbf{d}\}^T, \text{Im}\{\mathbf{d}\}^T]^T \in \mathbb{R}^{4Q}, \quad (8)$$

with

$$\boldsymbol{\omega}_1 = \boldsymbol{\theta}, \quad \boldsymbol{\omega}_2 = \boldsymbol{\phi}, \quad \boldsymbol{\omega}_3 = \text{Re}\{\mathbf{d}\}, \quad \boldsymbol{\omega}_4 = \text{Im}\{\mathbf{d}\}. \quad (9)$$

The angle-derivative steering matrices are

$$\dot{\mathbf{A}}_\theta \triangleq \left[\frac{\partial \mathbf{a}(\theta_q, \phi_q)}{\partial \theta_q} \right]_{q=1}^Q, \quad \dot{\mathbf{A}}_\phi \triangleq \left[\frac{\partial \mathbf{a}(\theta_q, \phi_q)}{\partial \phi_q} \right]_{q=1}^Q, \quad (10)$$

$$\dot{\mathbf{B}}_\theta \triangleq \left[\frac{\partial \mathbf{b}(\theta_q, \phi_q)}{\partial \theta_q} \right]_{q=1}^Q, \quad \dot{\mathbf{B}}_\phi \triangleq \left[\frac{\partial \mathbf{b}(\theta_q, \phi_q)}{\partial \phi_q} \right]_{q=1}^Q. \quad (11)$$

For a real-valued differentiable function $h(\mathbf{X})$ of a complex matrix, we use

$$dh(\mathbf{X}) = \text{Re}\left\{ \text{tr}\left[(\nabla_{\mathbf{X}} h(\mathbf{X}))^H d\mathbf{X} \right] \right\}. \quad (12)$$

II. DETAILED DERIVATION OF THE FISHER INFORMATION MATRIX

A. General FIM expression

Since the covariance $\sigma_s^2 \mathbf{I}_{L_r}$ of $\mathbf{y}_s[t]$ is independent of $\boldsymbol{\omega}$, the (i, j) -th FIM sub-block is [1]

$$\mathbf{J}_{\omega_i \omega_j} = \frac{2}{\sigma_s^2} \text{Re} \left\{ \sum_{t=1}^T \left(\frac{\partial \mathbf{u}_s[t]}{\partial \omega_i} \right)^H \frac{\partial \mathbf{u}_s[t]}{\partial \omega_j} \right\}. \quad (13)$$

Let $\mathbf{e}_q$ denote the q -th column of $\mathbf{I}_Q$. From (3),

$$\frac{\partial \mathbf{u}_s[t]}{\partial \theta_q} = \dot{\mathbf{B}}_\theta \mathbf{e}_q \mathbf{e}_q^T \mathbf{D} \mathbf{A}^H \mathbf{W} \mathbf{s}[t] + \mathbf{B} \mathbf{e}_q \mathbf{e}_q^T \mathbf{D} \dot{\mathbf{A}}_\theta^H \mathbf{W} \mathbf{s}[t], \quad (14)$$

$$\frac{\partial \mathbf{u}_s[t]}{\partial \phi_q} = \dot{\mathbf{B}}_\phi \mathbf{e}_q \mathbf{e}_q^T \mathbf{D} \mathbf{A}^H \mathbf{W} \mathbf{s}[t] + \mathbf{B} \mathbf{e}_q \mathbf{e}_q^T \mathbf{D} \dot{\mathbf{A}}_\phi^H \mathbf{W} \mathbf{s}[t], \quad (15)$$

$$\frac{\partial \mathbf{u}_s[t]}{\partial \text{Re}\{d_q\}} = \mathbf{B} \mathbf{e}_q \mathbf{e}_q^T \mathbf{A}^H \mathbf{W} \mathbf{s}[t], \quad (16)$$

$$\frac{\partial \mathbf{u}_s[t]}{\partial \text{Im}\{d_q\}} = j \mathbf{B} \mathbf{e}_q \mathbf{e}_q^T \mathbf{A}^H \mathbf{W} \mathbf{s}[t]. \quad (17)$$

Define

$$\mathbf{R}_c \triangleq \mathbf{W} \mathbf{W}^H. \quad (18)$$

Using $\mathbf{S}\mathbf{S}^H \simeq T\mathbf{I}_K$, the sum over snapshots in (13) depends on $\mathbf{W}$ only through $\mathbf{R}_c$.

B. Example: θ - θ block

Substituting (14) into (13) gives

$$\begin{aligned} [\mathbf{J}_{\theta\theta}]_{p,q} &= \frac{2T}{\sigma_s^2} \text{Re} \left\{ [\mathbf{D} \mathbf{A}^H \mathbf{R}_c \mathbf{A} \mathbf{D}^H]_{q,p} [\dot{\mathbf{B}}_\theta^H \dot{\mathbf{B}}_\theta]_{p,q} \right. \\ &\quad + [\mathbf{D} \mathbf{A}^H \mathbf{R}_c \dot{\mathbf{A}}_\theta \mathbf{D}^H]_{q,p} [\mathbf{B}^H \dot{\mathbf{B}}_\theta]_{p,q} \\ &\quad + [\mathbf{D} \dot{\mathbf{A}}_\theta^H \mathbf{R}_c \mathbf{A} \mathbf{D}^H]_{q,p} [\dot{\mathbf{B}}_\theta^H \mathbf{B}]_{p,q} \\ &\quad \left. + [\mathbf{D} \dot{\mathbf{A}}_\theta^H \mathbf{R}_c \dot{\mathbf{A}}_\theta \mathbf{D}^H]_{q,p} [\mathbf{B}^H \mathbf{B}]_{p,q} \right\}. \end{aligned} \quad (19)$$

Thus,

$$\mathbf{J}_{\theta\theta} = \frac{2T}{\sigma_s^2} \text{Re}\{\mathbf{J}_{11}\}, \quad (20)$$

with

$$\begin{aligned} \mathbf{J}_{11} &\triangleq (\mathbf{D} \mathbf{A}^H \mathbf{R}_c \mathbf{A} \mathbf{D}^H)^T \odot (\dot{\mathbf{B}}_\theta^H \dot{\mathbf{B}}_\theta) \\ &\quad + (\mathbf{D} \mathbf{A}^H \mathbf{R}_c \dot{\mathbf{A}}_\theta \mathbf{D}^H)^T \odot (\mathbf{B}^H \dot{\mathbf{B}}_\theta) \\ &\quad + (\mathbf{D} \dot{\mathbf{A}}_\theta^H \mathbf{R}_c \mathbf{A} \mathbf{D}^H)^T \odot (\dot{\mathbf{B}}_\theta^H \mathbf{B}) \\ &\quad + (\mathbf{D} \dot{\mathbf{A}}_\theta^H \mathbf{R}_c \dot{\mathbf{A}}_\theta \mathbf{D}^H)^T \odot (\mathbf{B}^H \mathbf{B}). \end{aligned} \quad (21)$$

C. Remaining FIM blocks

The remaining blocks follow analogously:

$$\begin{aligned} \mathbf{J}_{12} &\triangleq (\mathbf{D} \mathbf{A}^H \mathbf{R}_c \mathbf{A} \mathbf{D}^H)^T \odot (\dot{\mathbf{B}}_\theta^H \dot{\mathbf{B}}_\phi) \\ &\quad + (\mathbf{D} \mathbf{A}^H \mathbf{R}_c \dot{\mathbf{A}}_\theta \mathbf{D}^H)^T \odot (\mathbf{B}^H \dot{\mathbf{B}}_\phi) \\ &\quad + (\mathbf{D} \dot{\mathbf{A}}_\phi^H \mathbf{R}_c \mathbf{A} \mathbf{D}^H)^T \odot (\dot{\mathbf{B}}_\theta^H \mathbf{B}) \\ &\quad + (\mathbf{D} \dot{\mathbf{A}}_\phi^H \mathbf{R}_c \dot{\mathbf{A}}_\theta \mathbf{D}^H)^T \odot (\mathbf{B}^H \mathbf{B}), \\ \mathbf{J}_{22} &\triangleq (\mathbf{D} \mathbf{A}^H \mathbf{R}_c \mathbf{A} \mathbf{D}^H)^T \odot (\dot{\mathbf{B}}_\phi^H \dot{\mathbf{B}}_\phi) \end{aligned} \quad (22)$$

$$\begin{aligned}
& + (\mathbf{D}\mathbf{A}^H\mathbf{R}_c\dot{\mathbf{A}}_\phi\mathbf{D}^H)^T \odot (\mathbf{B}^H\dot{\mathbf{B}}_\phi) \\
& + (\mathbf{D}\dot{\mathbf{A}}_\phi^H\mathbf{R}_c\mathbf{A}\mathbf{D}^H)^T \odot (\dot{\mathbf{B}}_\phi^H\mathbf{B}) \\
& + (\mathbf{D}\dot{\mathbf{A}}_\phi^H\mathbf{R}_c\dot{\mathbf{A}}_\phi\mathbf{D}^H)^T \odot (\mathbf{B}^H\mathbf{B}), \quad (23)
\end{aligned}$$

$$\begin{aligned}
\mathbf{J}_{13} & \triangleq (\mathbf{A}^H\mathbf{R}_c\mathbf{A}\mathbf{D}^H)^T \odot (\dot{\mathbf{B}}_\theta^H\mathbf{B}) \\
& + (\mathbf{A}^H\mathbf{R}_c\dot{\mathbf{A}}_\theta\mathbf{D}^H)^T \odot (\mathbf{B}^H\mathbf{B}), \quad (24)
\end{aligned}$$

$$\begin{aligned}
\mathbf{J}_{23} & \triangleq (\mathbf{A}^H\mathbf{R}_c\mathbf{A}\mathbf{D}^H)^T \odot (\dot{\mathbf{B}}_\phi^H\mathbf{B}) \\
& + (\mathbf{A}^H\mathbf{R}_c\dot{\mathbf{A}}_\phi\mathbf{D}^H)^T \odot (\mathbf{B}^H\mathbf{B}), \quad (25)
\end{aligned}$$

$$\mathbf{J}_{33} \triangleq (\mathbf{A}^H\mathbf{R}_c\mathbf{A})^T \odot (\mathbf{B}^H\mathbf{B}). \quad (26)$$

The full FIM is

$$\mathbf{J} = \frac{2T}{\sigma_s^2} \begin{bmatrix} \text{Re}\{\mathbf{J}_{11}\} & \text{Re}\{\mathbf{J}_{12}\} & \text{Re}\{\mathbf{J}_{13}\} & -\text{Im}\{\mathbf{J}_{13}\} \\ \text{Re}\{\mathbf{J}_{12}^T\} & \text{Re}\{\mathbf{J}_{22}\} & \text{Re}\{\mathbf{J}_{23}\} & -\text{Im}\{\mathbf{J}_{23}\} \\ \text{Re}\{\mathbf{J}_{13}^T\} & \text{Re}\{\mathbf{J}_{23}^T\} & \text{Re}\{\mathbf{J}_{33}\} & -\text{Im}\{\mathbf{J}_{33}\} \\ -\text{Im}\{\mathbf{J}_{13}^T\} & -\text{Im}\{\mathbf{J}_{23}^T\} & -\text{Im}\{\mathbf{J}_{33}^T\} & \text{Re}\{\mathbf{J}_{33}\} \end{bmatrix}. \quad (27)$$

III. REDUCED-DOMAIN FIM

The letter uses

$$\mathbf{N} \triangleq [\mathbf{G}, \mathbf{A}, \dot{\mathbf{A}}_\theta, \dot{\mathbf{A}}_\phi] \in \mathbb{C}^{L_t \times (K+3Q)} \quad (28)$$

and its compact QR factorization

$$\mathbf{N} = \mathbf{U}\mathbf{R}, \quad \mathbf{U}^H\mathbf{U} = \mathbf{I}_{L_d}, \quad L_d = \text{rank}(\mathbf{N}). \quad (29)$$

Writing

$$\mathbf{W} = \mathbf{U}\mathbf{X} \quad (30)$$

gives

$$\mathbf{R}_c = \mathbf{U}\mathbf{X}\mathbf{X}^H\mathbf{U}^H. \quad (31)$$

Define

$$\bar{\mathbf{G}} \triangleq \mathbf{U}^H\mathbf{G}, \quad \bar{\mathbf{A}} \triangleq \mathbf{U}^H\mathbf{A}, \quad (32)$$

$$\dot{\bar{\mathbf{A}}}_\theta \triangleq \mathbf{U}^H\dot{\mathbf{A}}_\theta, \quad \dot{\bar{\mathbf{A}}}_\phi \triangleq \mathbf{U}^H\dot{\mathbf{A}}_\phi. \quad (33)$$

Then, e.g.,

$$\mathbf{A}^H\mathbf{R}_c\mathbf{A} = \bar{\mathbf{A}}^H\mathbf{X}\mathbf{X}^H\bar{\mathbf{A}}, \quad (34)$$

with analogous identities for all mixed steering-derivative terms. Hence the FIM can be evaluated entirely in the L_d -dimensional domain.

IV. GRADIENT OF THE SUM RATE IN THE REDUCED DOMAIN

The reduced-dimensional channel of user k is $\bar{\mathbf{g}}_k = \mathbf{U}^H\mathbf{g}_k$, and we write $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_K]$. For notational convenience, define

$$S_k \triangleq |\bar{\mathbf{g}}_k^H\mathbf{x}_k|^2, \quad I_k \triangleq \sum_{j \neq k} |\bar{\mathbf{g}}_k^H\mathbf{x}_j|^2 + \sigma_{ck}^2,$$

$$T_k \triangleq S_k + I_k, \quad \eta_k \triangleq \frac{\bar{\mathbf{g}}_k^H\mathbf{x}_k}{I_k}, \quad \beta_k \triangleq \frac{S_k}{I_k T_k}.$$

Furthermore, let $\mathbf{\Lambda}_1 \triangleq \text{diag}\{\eta_1, \dots, \eta_K\}$ and $\mathbf{\Lambda}_2 \triangleq \text{diag}\{\beta_1, \dots, \beta_K\}$.

The achievable rate of user k can be written as

$$r_k = \log \left(1 + \frac{S_k}{I_k} \right) = \log T_k - \log I_k.$$

Using

$$d|\bar{\mathbf{g}}_k^H\mathbf{x}_j|^2 = 2 \text{Re} \{ (\bar{\mathbf{g}}_k^H\mathbf{x}_j)^* \bar{\mathbf{g}}_k^H d\mathbf{x}_j \},$$

we have

$$dT_k = 2 \text{Re} \left\{ \sum_{j=1}^K (\bar{\mathbf{g}}_k^H\mathbf{x}_j)^* \bar{\mathbf{g}}_k^H d\mathbf{x}_j \right\},$$

and

$$dI_k = 2 \text{Re} \left\{ \sum_{j \neq k} (\bar{\mathbf{g}}_k^H\mathbf{x}_j)^* \bar{\mathbf{g}}_k^H d\mathbf{x}_j \right\}.$$

Therefore,

$$\begin{aligned}
dr_k &= \frac{dT_k}{T_k} - \frac{dI_k}{I_k} \\
&= 2 \text{Re} \left\{ \frac{(\bar{\mathbf{g}}_k^H\mathbf{x}_k)^*}{T_k} \bar{\mathbf{g}}_k^H d\mathbf{x}_k \right\} \\
&\quad - 2 \text{Re} \left\{ \sum_{j \neq k} \frac{S_k}{I_k T_k} (\bar{\mathbf{g}}_k^H\mathbf{x}_j)^* \bar{\mathbf{g}}_k^H d\mathbf{x}_j \right\}, \quad (35)
\end{aligned}$$

where we have used

$$\frac{1}{T_k} - \frac{1}{I_k} = -\frac{S_k}{I_k T_k} = -\beta_k.$$

For the desired-stream term, note that

$$\frac{\bar{\mathbf{g}}_k^H\mathbf{x}_k}{T_k} = \frac{\bar{\mathbf{g}}_k^H\mathbf{x}_k}{I_k} - \frac{S_k}{I_k T_k} \bar{\mathbf{g}}_k^H\mathbf{x}_k = \eta_k - \beta_k \bar{\mathbf{g}}_k^H\mathbf{x}_k.$$

Hence, (35) can be equivalently expressed as

$$dr_k = 2 \text{Re} \left\{ \eta_k^* \bar{\mathbf{g}}_k^H d\mathbf{x}_k - \sum_{j=1}^K \beta_k (\bar{\mathbf{g}}_k^H\mathbf{x}_j)^* \bar{\mathbf{g}}_k^H d\mathbf{x}_j \right\}. \quad (36)$$

Summing over all users gives

$$\begin{aligned}
d \sum_{k=1}^K r_k &= 2 \text{Re} \left\{ \sum_{k=1}^K \eta_k^* \bar{\mathbf{g}}_k^H d\mathbf{x}_k \right. \\
&\quad \left. - \sum_{j=1}^K \sum_{k=1}^K \beta_k (\bar{\mathbf{g}}_k^H\mathbf{x}_j)^* \bar{\mathbf{g}}_k^H d\mathbf{x}_j \right\}. \quad (37)
\end{aligned}$$

Defining $\bar{\mathbf{G}} \triangleq [\bar{\mathbf{g}}_1, \dots, \bar{\mathbf{g}}_K]$, the above differential can be written in matrix form as

$$d \sum_{k=1}^K r_k = \text{Re} \left\{ \text{tr} \left[(2\bar{\mathbf{G}}\mathbf{\Lambda}_1 - 2\bar{\mathbf{G}}\mathbf{\Lambda}_2\bar{\mathbf{G}}^H\mathbf{X})^H d\mathbf{X} \right] \right\}.$$

Therefore, under the convention $df = \Re\{\text{tr}[(\nabla_{\mathbf{X}}f)^H d\mathbf{X}]\}$, the gradient of the sum rate is

$$\nabla_{\mathbf{X}} \sum_{k=1}^K r_k = 2\bar{\mathbf{G}}\mathbf{\Lambda}_1 - 2\bar{\mathbf{G}}\mathbf{\Lambda}_2\bar{\mathbf{G}}^H\mathbf{X}. \quad (38)$$

V. GRADIENT OF THE CRLB METRIC IN THE REDUCED DOMAIN

A. Differential of the inverse FIM

Define

$$f_s(\mathbf{X}) \triangleq -\text{tr}(\mathbf{J}^{-1}). \quad (39)$$

Using

$$d(\mathbf{J}^{-1}) = -\mathbf{J}^{-1}(d\mathbf{J})\mathbf{J}^{-1}, \quad (40)$$

we have

$$df_s = -d \text{tr}(\mathbf{J}^{-1}) = \text{tr}(\mathbf{J}^{-1}(d\mathbf{J})\mathbf{J}^{-1}) = \text{tr}(\Phi d\mathbf{J}), \quad (41)$$

where $\Phi \triangleq \mathbf{J}^{-2}$. Note that Φ appears with a positive sign because $f_s(\mathbf{X}) = -\text{tr}(\mathbf{J}^{-1})$. Partition

$$\Phi = \begin{bmatrix} \Phi_{11} & \Phi_{12} & \Phi_{13} & \Phi_{14} \\ \Phi_{12}^T & \Phi_{22} & \Phi_{23} & \Phi_{24} \\ \Phi_{13}^T & \Phi_{23}^T & \Phi_{33} & \Phi_{34} \\ \Phi_{14}^T & \Phi_{24}^T & \Phi_{34}^T & \Phi_{44} \end{bmatrix}. \quad (42)$$

Let $c_s \triangleq 2T/\sigma_s^2$. Since $\mathbf{J}$ and Φ are real symmetric,

$$\begin{aligned} df_s = c_s \text{Re} \left\{ \text{tr} \left[\Phi_{11}^T d\mathbf{J}_{11} + 2\Phi_{12}^T d\mathbf{J}_{12} \right. \right. \\ + \Phi_{22}^T d\mathbf{J}_{22} + 2(\Phi_{13}^T + j\Phi_{14}^T) d\mathbf{J}_{13} \\ + 2(\Phi_{23}^T + j\Phi_{24}^T) d\mathbf{J}_{23} \\ \left. \left. + (\Phi_{33}^T + \Phi_{44}^T + 2j\Phi_{34}^T) d\mathbf{J}_{33} \right] \right\}. \end{aligned} \quad (43)$$

B. Compact differential

For compactness, define

$$\begin{aligned} \bar{\mathbf{A}}_o &\triangleq \bar{\mathbf{A}}\mathbf{D}^H, \\ \bar{\mathbf{A}}_\theta &\triangleq \dot{\bar{\mathbf{A}}}_\theta \mathbf{D}^H, \quad \bar{\mathbf{A}}_\phi &\triangleq \dot{\bar{\mathbf{A}}}_\phi \mathbf{D}^H. \end{aligned} \quad (44)$$

and

$$\begin{aligned} \mathbf{B}_{oo} &\triangleq \mathbf{B}^H \mathbf{B}, & \mathbf{B}_{o\theta} &\triangleq \mathbf{B}^H \dot{\mathbf{B}}_\theta, \\ \mathbf{B}_{o\phi} &\triangleq \mathbf{B}^H \dot{\mathbf{B}}_\phi, & \mathbf{B}_{\theta o} &\triangleq \dot{\mathbf{B}}_\theta^H \mathbf{B}, \\ \mathbf{B}_{\theta\theta} &\triangleq \dot{\mathbf{B}}_\theta^H \dot{\mathbf{B}}_\theta, & \mathbf{B}_{\theta\phi} &\triangleq \dot{\mathbf{B}}_\theta^H \dot{\mathbf{B}}_\phi, \\ \mathbf{B}_{\phi o} &\triangleq \dot{\mathbf{B}}_\phi^H \mathbf{B}, & \mathbf{B}_{\phi\phi} &\triangleq \dot{\mathbf{B}}_\phi^H \dot{\mathbf{B}}_\phi. \end{aligned} \quad (45)$$

The identity

$$\begin{aligned} \text{Re} \left\{ \text{tr} \left(\mathbf{Z}^T d[(\mathbf{C}^H \mathbf{R} \mathbf{D})^T \odot \mathbf{E}] \right) \right\} \\ = \text{Re} \left\{ \text{tr} \left(d\mathbf{R} \mathbf{D} (\mathbf{Z} \odot \mathbf{E}) \mathbf{C}^H \right) \right\}. \end{aligned} \quad (46)$$

converts each term of (43) into a differential with respect to $\mathbf{R}_\mathbf{X} = \mathbf{X}\mathbf{X}^H$. Therefore,

$$df_s = c_s \text{Re} \left\{ \text{tr} [d(\mathbf{X}\mathbf{X}^H) \mathbf{Q}_e] \right\}, \quad (47)$$

where

$$\mathbf{Q}_e = \mathbf{Q}_{11} + \mathbf{Q}_{12} + \mathbf{Q}_{22} + \mathbf{Q}_{13} + \mathbf{Q}_{23} + \mathbf{Q}_{33}, \quad (48)$$

and

$$\begin{aligned} \mathbf{Q}_{11} &= \bar{\mathbf{A}}_o (\Phi_{11} \odot \mathbf{B}_{\theta\theta}) \bar{\mathbf{A}}_o^H \\ &+ \bar{\mathbf{A}}_\theta (\Phi_{11} \odot \mathbf{B}_{o\theta}) \bar{\mathbf{A}}_o^H \\ &+ \bar{\mathbf{A}}_o (\Phi_{11} \odot \mathbf{B}_{\theta o}) \bar{\mathbf{A}}_\theta^H \\ &+ \bar{\mathbf{A}}_\theta (\Phi_{11} \odot \mathbf{B}_{oo}) \bar{\mathbf{A}}_\theta^H, \end{aligned} \quad (49)$$

$$\begin{aligned} \mathbf{Q}_{12} &= 2\bar{\mathbf{A}}_o (\Phi_{12} \odot \mathbf{B}_{\theta\phi}) \bar{\mathbf{A}}_o^H \\ &+ 2\bar{\mathbf{A}}_\theta (\Phi_{12} \odot \mathbf{B}_{o\phi}) \bar{\mathbf{A}}_o^H \\ &+ 2\bar{\mathbf{A}}_o (\Phi_{12} \odot \mathbf{B}_{\theta o}) \bar{\mathbf{A}}_\phi^H \\ &+ 2\bar{\mathbf{A}}_\theta (\Phi_{12} \odot \mathbf{B}_{oo}) \bar{\mathbf{A}}_\phi^H, \end{aligned} \quad (50)$$

$$\begin{aligned} \mathbf{Q}_{22} &= \bar{\mathbf{A}}_o (\Phi_{22} \odot \mathbf{B}_{\phi\phi}) \bar{\mathbf{A}}_o^H \\ &+ \bar{\mathbf{A}}_\phi (\Phi_{22} \odot \mathbf{B}_{o\phi}) \bar{\mathbf{A}}_o^H \\ &+ \bar{\mathbf{A}}_o (\Phi_{22} \odot \mathbf{B}_{\phi o}) \bar{\mathbf{A}}_\phi^H \\ &+ \bar{\mathbf{A}}_\phi (\Phi_{22} \odot \mathbf{B}_{oo}) \bar{\mathbf{A}}_\phi^H, \end{aligned} \quad (51)$$

$$\begin{aligned} \mathbf{Q}_{13} &= \bar{\mathbf{A}}_o [(2\Phi_{13} + 2j\Phi_{14}) \odot \mathbf{B}_{\theta o}] \bar{\mathbf{A}}^H \\ &+ \bar{\mathbf{A}}_\theta [(2\Phi_{13} + 2j\Phi_{14}) \odot \mathbf{B}_{oo}] \bar{\mathbf{A}}^H, \end{aligned} \quad (52)$$

$$\begin{aligned} \mathbf{Q}_{23} &= \bar{\mathbf{A}}_o [(2\Phi_{23} + 2j\Phi_{24}) \odot \mathbf{B}_{\phi o}] \bar{\mathbf{A}}^H \\ &+ \bar{\mathbf{A}}_\phi [(2\Phi_{23} + 2j\Phi_{24}) \odot \mathbf{B}_{oo}] \bar{\mathbf{A}}^H, \end{aligned} \quad (53)$$

$$\mathbf{Q}_{33} = \bar{\mathbf{A}} [(2\Phi_{33} + \Phi_{44} + 2j\Phi_{34}) \odot \mathbf{B}_{oo}] \bar{\mathbf{A}}^H. \quad (54)$$

Using $d(\mathbf{X}\mathbf{X}^H) = d\mathbf{X}\mathbf{X}^H + \mathbf{X}d\mathbf{X}^H$ gives

$$\begin{aligned} df_s &= c_s \text{Re} \left\{ \text{tr} [(d\mathbf{X}\mathbf{X}^H + \mathbf{X}d\mathbf{X}^H) \mathbf{Q}_e] \right\} \\ &= c_s \text{Re} \left\{ \text{tr} [(\mathbf{Q}_e^H \mathbf{X})^H d\mathbf{X}] + \text{tr} [(\mathbf{Q}_e \mathbf{X})^H d\mathbf{X}] \right\} \\ &= c_s \text{Re} \left\{ \text{tr} [((\mathbf{Q}_e + \mathbf{Q}_e^H) \mathbf{X})^H d\mathbf{X}] \right\}. \end{aligned} \quad (55)$$

Comparing (55) with the differential convention in (12) gives

$$\nabla_{\mathbf{X}} [-\text{tr}(\mathbf{J}^{-1})] = c_s (\mathbf{Q}_e + \mathbf{Q}_e^H) \mathbf{X}. \quad (56)$$

Equivalently,

$$\nabla_{\mathbf{X}} \text{tr}(\mathbf{J}^{-1}) = -(\mathbf{M} + \mathbf{M}^H) \mathbf{X}, \quad \mathbf{M} \triangleq c_s \mathbf{Q}_e. \quad (57)$$

REFERENCES

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